Table 3.2 Salient Features of Static and Kinetic Friction

Static friction	Kinetic friction
It opposes the starting of motion	It opposes the relative motion of the object with respect to the surface
Independent of surface area of contact	Independent of surface area of contact
μ_s depends on the nature of materials in mutual contact	μ_k depends on nature of materials and temperature of the surface
Depends on the magnitude of applied force	Independent of magnitude of applied force
It can take values from zero to $\mu_s N$	It can never be zero and always equals to $\mu_k N$ whatever be the speed (true v <10 ms^{-1})
$f_s^{max} > f_k$	It is less than maximum value of static friction
$\mu_s > \mu_k$	Coefficient of kinetic friction is less than coefficient of static friction

The variation of both static and kinetic frictional forces with external applied force is graphically shown in Figure 3.25.

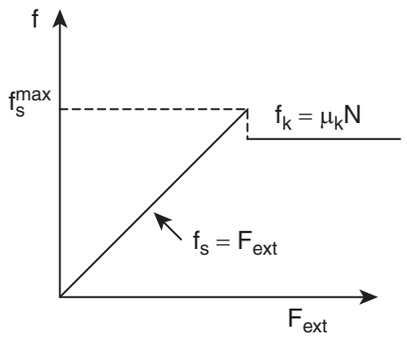


Figure 3.25 Variation of static and kinetic frictional forces with external applied force

The Figure 3.25 shows that static friction increases linearly with external applied force till it reaches the maximum. If the object begins to move then the kinetic friction is slightly lesser than the maximum static friction. Note that the kinetic friction is constant and it is independent of applied force.

Note

The relation $f_s = \mu_s N$ is not a vector relation. This is because the normal force N and f_s are not in the same direction even though f_s is equal to μ_s times the normal force. This is also true in the case of kinetic friction.

3.6.4 To Move an Object - Push or pull? Which is easier?

When a body is pushed at an arbitrary angle θ $\left(0 \text{ to } \frac{\pi}{2}\right)$, the applied force F can be resolved into two components as F sinθ parallel to the surface and F cosθ perpendicular to the surface as shown in Figure 3.26. The total downward force acting on the body is mg + Fcosθ. It implies *that* the normal force acting on the body increases. Since there is no acceleration along the vertical direction the normal force N is equal to

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